

pamaldi at SemEval-2026 Task 11: Neuro-Symbolic Syllogistic Reasoning via LLM-Guided Structure Extraction and Deterministic Validation

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Abstract

We describe our participation in SemEval-2026 Task 11, Subtask 1: determining the formal validity of syllogisms in English while minimizing the influence of content plausibility. Our system implements a neuro-symbolic pipeline that strictly separates the neural and symbolic components of syllogistic reasoning. An LLM extracts the formal structure of natural-language syllogisms—identifying the proposition types (A, E, I, O) and the three terms—while the syllogistic figure is computed deterministically and a symbolic validator checks whether the resulting mood-figure pair belongs to the 24 classically valid Aristotelian forms. Self-consistency voting across multiple extraction samples and targeted post-processing rules further improve robustness. On the official evaluation, our system achieves 96.34% accuracy, a Total Content Effect (TCE) of 1.02, and a combined score of 56.57. On the symbolic path (96.3% of instances), the validator is content-agnostic by construction; residual content sensitivity arises only when extraction errors map instances onto forms whose validity happens to align with plausibility. All seven residual errors are attributable to neural components: four stem from structured-extraction failures in the symbolic path and three from direct LLM fallback judgments. Compared to pure-LLM baselines on the same backbone, our system more than doubles the combined score (from 26.52 to 56.57) and reduces TCE by nearly an order of magnitude. Swapping the extractor to a second backbone (Claude Sonnet 4.5) preserves combined score and TCE, confirming that content-invariance is contributed by the symbolic stage rather than any particular LLM.

1 Introduction

Large language models (LLMs) exhibit a persistent vulnerability to *content effects* in logical reasoning: they tend to accept logically invalid arguments when the conclusion is plausible, and reject valid

arguments when the conclusion seems implausible (Dasgupta et al., 2022; Eisape et al., 2024). This entanglement of formal reasoning with world knowledge undermines the reliability of LLMs in tasks requiring strict logical inference.

SemEval-2026 Task 11 (Valentino et al., 2026) addresses this challenge by evaluating systems on syllogistic reasoning across four subtasks of increasing complexity, from English-only binary classification to multilingual settings with irrelevant premises. In this paper, we focus on **Subtask 1**: predicting the formal validity of categorical syllogisms in English, where the official ranking metric jointly rewards high accuracy and penalizes content bias through the Total Content Effect (TCE).

Our key insight is that syllogistic validity is a purely structural property: it depends entirely on the *mood* (the combination of proposition types A, E, I, O) and the *figure* (the arrangement of terms across premises), not on the truth or plausibility of the premises. This motivates a neuro-symbolic architecture that decomposes the problem into two stages: (1) a *neural extraction* stage, where an LLM parses the natural-language syllogism into its formal components, and (2) a *symbolic validation* stage, where a deterministic checker verifies whether the extracted mood-figure pair belongs to the 24 classically valid Aristotelian forms.

Our contributions are: (1) a strict neuro-symbolic decomposition in which the LLM is forbidden from producing reasoning steps and emits only four AEIO labels plus three term strings, with all inference performed by a closed-form lookup; (2) a controlled comparison, on the same backbone, against direct-prompt, chain-of-thought, self-consistent chain-of-thought, and an LLM-generates-Prolog variant, showing that reducing the LLM’s reasoning role improves both accuracy and TCE, and that sampling alone does not close the content-effect gap; (3) a branch-level decomposition showing that the symbolic validator con-

tributes no content-dependent errors, while neural components (both the extractor and the fallback) account for all observed content sensitivity; and (4) a multi-model study of the extractor demonstrating that the design claim—that the symbolic stage is the invariant content-immune component—holds across backbones.

Our submitted system achieves a combined score of 56.57 with 96.34% accuracy and a TCE of only 1.02. Code, prompts, and evaluation scripts are publicly available.^{1,2}

2 Background

2.1 Task Description

SemEval-2026 Task 11 (Valentino et al., 2026) investigates whether computational systems can perform formal logical reasoning independently of content plausibility. Subtask 1 requires predicting the binary validity label of syllogisms presented in English. Each syllogism consists of two premises and a conclusion, and is annotated with both a validity label (the target) and a plausibility label (reflecting real-world commonsense). The training set contains 200 instances and the test set 191 instances, each balanced across four conditions: valid-plausible (VP), valid-implausible (VI), invalid-plausible (IP), and invalid-implausible (II).

2.2 Evaluation Metrics

Systems are evaluated on three metrics: (1) **Overall Accuracy** (ACC); (2) **Total Content Effect** (TCE), a composite score measuring the average accuracy difference due to plausibility across logical conditions—a lower TCE indicates greater robustness to content bias; and (3) the **Primary Ranking Metric**, computed as

$$\text{Score} = \frac{\text{ACC}}{1 + \ln(1 + \text{TCE})}. \quad (1)$$

This metric rewards high accuracy while smoothly penalizing content bias.

2.3 Categorical Syllogisms

A categorical syllogism consists of exactly three terms—the *major term* (P), *minor term* (S), and *middle term* (M)—distributed across two premises and a conclusion. Each proposition takes one of four forms: **A** (“All S are P”), **E** (“No S are P”),

I (“Some S are P”), or **O** (“Some S are not P”). The *mood* is the ordered triple of proposition types (e.g. AAA, EIO); the *figure* (1–4) is determined by the positions of the middle term across the two premises (Copi et al., 2014). Under the traditional Aristotelian interpretation with *existential import*, exactly 24 mood-figure pairs are valid, distributed across the four figures.

2.4 Related Work

Content effects in LLM reasoning have been extensively documented (Dasgupta et al., 2022; Eisape et al., 2024; Bertolazzi et al., 2024; Ozeki et al., 2024; Seals and Shalin, 2024), with mechanistic analyses of syllogistic inference circuits (Kim et al., 2025). Mitigation approaches include fine-grained activation steering (Valentino et al., 2025; Maraia et al., 2026), quasi-symbolic chain-of-thought (Ranaldi et al., 2025), faithful-CoT methods (Lyu et al., 2023; Xu et al., 2024), and LLM-symbolic theorem-prover combinations (Quan et al., 2024).

Our approach differs in a stricter sense. Whereas faithful-CoT and symbolic-CoT methods (Lyu et al., 2023; Xu et al., 2024; Ranaldi et al., 2025) and theorem-prover hybrids (Quan et al., 2024) still rely on LLM-generated reasoning steps that are validated or executed downstream, our pipeline forbids the LLM from producing any reasoning: it emits only AEIO type labels and term strings. The syllogistic figure is computed deterministically from term positions, and all inference is a closed-form lookup against the 24 valid Aristotelian forms. As we show in Section 3.1, this stricter separation yields both higher accuracy and lower TCE than an LLM-generates-Prolog variant on the same data.

3 System Overview

3.1 Design Evolution

The final system emerged from an iterative exploration of four architectures, each addressing limitations uncovered in the previous one. We re-executed all approaches on the official test set (191 instances) after gold labels were released; results are summarized in Table 1.

Prolog Code Generation with Reflexion. Our first approach prompted the LLM to generate executable SWI-Prolog programs encoding each syllogism’s logical structure, with a Reflexion self-correction loop (Shinn et al., 2023) providing up to three retry attempts on execution failure. While

¹https://github.com/pamaldi/pamaldi_semeval_task11

²Language of data: English.

conceptually elegant, this approach achieved only 90.58% accuracy with a TCE of 8.49 (combined score 27.87). Only 110 of 191 test instances succeeded on the first attempt, and the LLM could encode content bias directly into the generated rules.

Neuro-Symbolic with Full LLM Extraction.

Observing that *validity checking does not require code generation*, we redesigned the pipeline as a two-stage system: the LLM extracts syllogistic structure (types, terms, figure) as a pure NLU task, while a deterministic module checks membership in the 24 valid Aristotelian forms. With self-consistency voting ($k = 4$), extraction reflexion, and figure verification, this variant reached 97.91% accuracy and TCE of 2.13 (combined score 45.74). Error analysis revealed two bottlenecks: incorrect figure computation by the LLM, and I/O proposition type confusion.

Simplified Extraction with Deterministic Figure.

To eliminate figure errors—the most frequent category—we further reduced the LLM’s role: it extracts *only* proposition types and terms, while the figure is computed *deterministically* from the middle term’s position. With $k = 3$ samples and no reflexion, this reached 97.38% accuracy and a lower TCE of 2.08 (combined score 45.81).

Incremental Meta-Learning. We also experimented with automatic prompt refinement: training errors were analyzed every 50 samples and used to generate an enhanced extraction prompt with few-shot examples and guidelines (the prompt grew from 15,878 to 24,585 characters across four checkpoints). While this helped on training data, it reached only 95.29% accuracy on the test set with a TCE of 6.25 (combined score 31.97), suggesting the meta-learned prompt overfit to training error patterns.

Convergence. The submitted system combines the most effective elements: simplified NLU extraction, deterministic figure computation, self-consistency voting, targeted post-processing rules, and an LLM fallback. The central finding across all approaches is that errors consistently originate in neural extraction—never in deterministic validation—and that more deterministic components yield lower content effects.

3.2 Final Pipeline

The submitted system follows a two-stage neuro-symbolic pipeline: (1) the LLM extracts the syl-

Approach	Acc.	TCE	Score
Prolog + Reflexion	90.58	8.49	27.87
Incr. Meta-Learning	95.29	6.25	31.97
Full LLM Extraction	97.91	2.13	45.74
Simpl. Det. Figure	97.38	2.08	45.81
Final (submitted)	96.34	1.02	56.57

Table 1: Test set results (191 instances) across explored approaches. The final submitted system achieves the best combined score thanks to its lowest TCE. The first four rows are post-hoc re-evaluations on the official test set.

Figure	Valid Moods
1	AAA, EAE, AII, EIO, AAI, EAO
2	EAE, AEE, EIO, AOO, EAO, AEO
3	AAI, IAI, AII, EAO, OAO, EIO
4	AAI, AEE, IAI, EAO, EIO, AEO

Table 2: The 24 valid syllogistic forms under Aristotelian logic with existential import, by figure.

logism’s proposition types and terms as three independent samples, aggregated by majority vote; (2) post-processing rules correct common errors (I/O confusion, double negation, compound terms); (3) the syllogistic figure is computed deterministically from the middle term’s position; (4) a symbolic validator checks membership in the 24 valid Aristotelian forms; (5) when structured extraction fails, a direct LLM fallback provides a validity judgment.

3.3 Neural Extraction Stage

Given a syllogism in free text, the LLM extracts the **three terms** (S, P, M) and the **proposition types** (A, E, I, O) for each of the two premises and the conclusion. The prompt includes detailed I/O distinction examples and a worked example for term identification (full template in Appendix A).

Self-Consistency Voting. We sample $k = 3$ independent extractions at temperatures 0.1, 0.3, and 0.5, selecting the most frequent mood–term combination. When all three disagree, the first extraction is used.

3.4 Symbolic Validation Stage

The deterministic checker verifies whether the extracted mood–figure pair belongs to the 24 valid Aristotelian forms (Table 2). This component contains no learned parameters and makes no reference to semantic content, ensuring complete immunity to content effects by construction.

3.5 Post-Processing and Fallback

Between extraction and validation, rule-based post-processing corrects systematic errors: **I/O disambiguation** detects negation markers to fix $I \leftrightarrow O$ misclassifications; **double-negation handling** recognizes patterns like “no X are not Y” as type A; **compound-term normalization** ensures consistent matching of complex noun phrases. When structured extraction fails—i.e. none of the $k = 3$ samples yields a well-formed mood-figure pair (3.7% of test instances, 7/191)—the system falls back to a direct LLM validity judgment. As we show in Section 5.2, the fallback path is a second source of content-dependent errors in addition to the extraction stage of the symbolic path.

4 Experimental Setup

Data and Model. Subtask 1 provides 200 training and 191 test English syllogisms, balanced across four validity-plausibility conditions. We use the training set for prompt development and error analysis only; no parameters are fine-tuned. The extraction and fallback components use Qwen3-32B³ with few-shot prompting. For self-consistency, we sample $k = 3$ extractions at temperatures 0.1, 0.3, and 0.5. The pipeline is implemented in Python 3.11 using boto3 for Bedrock inference and the Anthropic Python SDK for the multi-model study (Section 5.3); all other components are pure Python.

Baselines. To isolate the contribution of the neuro-symbolic decomposition, we compare against three pure-LLM baselines on the same backbone (Qwen3-32B) and the same 191-instance test set: **B1: Direct prompt** (“Is the following syllogism valid? Answer VALID or INVALID”, temperature 0, single sample); **B2: Chain-of-thought** (same prompt plus “Think step by step, then on the last line answer only VALID or INVALID”, temperature 0.3, single sample, max 512 tokens); **B3: CoT + self-consistency** ($k = 4$ CoT samples at temperatures [0.0, 0.3, 0.5, 0.7], majority vote with tie-break by the temperature-0 sample). To probe cross-model generalisation of the extractor, we also run the full pipeline with the symbolic validator held fixed and an alternative backbone (Section 5.3; Appendix D details model IDs).

Development. The system was developed through iterative error analysis on the training set.

³Accessed via Amazon Bedrock.

	Plausible	Implausible
Valid	95.83% (46/48)	95.83% (46/48)
Invalid	97.87% (46/47)	95.83% (46/48)

Table 3: Accuracy across validity-plausibility subgroups (test set). Near-uniform performance yields TCE = 1.02; overall accuracy 96.34%, combined score 56.57.

System (Qwen3-32B)	Acc.	TCE	Score
B1: Direct prompt	79.58	22.16	19.21
B2: Chain-of-thought	89.01	9.55	26.52
B3: CoT + SC ($k = 4$)	89.53	10.59	25.95
Prolog + Reflexion	90.58	8.49	27.87
Ours (final)	96.34	1.02	56.57

Table 4: Controlled comparison on the official test set (191 instances), same Qwen3-32B backbone. B1–B3 are pure-LLM baselines; the Prolog row uses the LLM for code generation followed by symbolic execution. All pure-LLM variants exhibit substantial content effect ($TCE \geq 8$), and self-consistency sampling (B3) does not close the gap left by chain-of-thought (B2).

Key iterations included adding I/O disambiguation rules, restoring the full 24 Aristotelian forms (with existential import) after confirming that the task adopts the traditional interpretation, and fixing self-consistency voting and fallback parsing bugs.

5 Results

Our system achieves 96.34% accuracy (184/191), a TCE of 1.02, and a combined score of 56.57 on the official evaluation. Table 3 reports accuracy across the four validity-plausibility subgroups: performance is near-identical across all four conditions, confirming that the symbolic validation stage successfully decouples validity judgments from content plausibility.

The errors comprise 4 false negatives and 3 false positives, spread across subgroups as (VP=2, VI=2, IP=1, II=2).

5.1 Comparison with Pure-LLM Baselines

Table 4 compares our system against three pure-LLM baselines on the same 191-instance test set, using the same Qwen3-32B backbone. The progression across B1–B3 yields four findings.

Direct prompting exhibits severe content effect. B1 reaches 79.58% overall accuracy but TCE of 22.16. Accuracy varies from 97.92% on invalid-implausible instances (where plausibility and validity align) down to 64.58% on valid-implausible instances, reproducing the asymmetric content bias

documented by Dasgupta et al. (2022) and Eisape et al. (2024): the model systematically rejects valid syllogisms whose conclusions sound implausible.

Chain-of-thought helps, but incompletely. B2 lifts overall accuracy to 89.01% and roughly halves TCE to 9.55; the valid–implausible subgroup improves from 64.58% to 89.58%, showing that explicit reasoning helps the model override plausibility-congruent rejection. However, a different asymmetry emerges: B2’s weakest subgroup becomes invalid–plausible (82.98%). The direction of content bias flips, but its magnitude remains substantial.

Self-consistency does not close the remaining gap. B3 adds self-consistency voting across $k = 4$ CoT samples at temperatures $[0.0, 0.3, 0.5, 0.7]$. Despite four times the compute, B3’s TCE (10.59) is statistically indistinguishable from B2’s (9.55), and its combined score (25.95) is marginally worse. The invalid–plausible subgroup remains at 82.98% in both B2 and B3: majority voting across samples that share the same plausibility-driven bias cannot cancel that bias. This confirms that more sampling is not sufficient—closing the content-effect gap requires a structural solution.

The neuro-symbolic decomposition closes the gap. Our system reaches a combined score of 56.57, more than double the best pure-LLM baseline (26.52). Across the four subgroups it ranges only from 95.83% to 97.87%, in contrast to pure-LLM baselines whose subgroup accuracies span 15–33 percentage points. Since all baselines and our system share the same Qwen3-32B backbone, this gap cannot be attributed to the model’s reasoning capacity: it is a direct consequence of forbidding the LLM from making validity judgments and delegating all inference to the deterministic validator.

5.2 Fallback Branch Analysis

Of the 191 test instances, 184 are resolved via the symbolic path and 7 via the LLM fallback. Table 5 decomposes the system’s accuracy and TCE by branch. The symbolic path, which handles 96.3% of instances, achieves 97.83% accuracy with a TCE of 2.13. The fallback, invoked on only 7 instances, achieves 57.14% accuracy; its TCE happens to be 0.00 because its three errors fall in subgroups where the small denominators yield no accuracy gap.

Path	N	Acc.	TCE	Score
Symbolic only	184	97.83	2.13	45.69
Fallback only	7	57.14	0.00	57.14
System (union)	191	96.34	1.02	56.57

Table 5: Per-branch decomposition on the test set. The symbolic validator introduces no errors on correctly extracted structures; residual content sensitivity in the symbolic path arises only when extraction errors map an instance onto a form whose validity happens to align with plausibility. The fallback’s TCE of 0.00 reflects the specific subgroup composition of the 7 fallback instances rather than a structural property. The system-level TCE of 1.02 is lower than the symbolic path’s 2.13 because the two paths contribute to different subgroup mixes.

Extractor	Acc.	TCE	Score	FB%
Qwen3-32B (submitted)	96.34	1.02	56.57	3.7
Claude Sonnet 4.5	97.38	1.06	56.47	1.0

Table 6: Swapping the extractor LLM while holding the symbolic validator, post-processing, and fallback handler fixed. Combined score and TCE are stable across the two backbones, while the stronger extractor reduces the fallback rate. FB% is the fraction of test instances routed to the LLM fallback branch.

The low system-level TCE of 1.02 is an averaging effect across branches with different subgroup compositions rather than a property of either path in isolation: the symbolic validator is content-invariant *by construction*, but residual content sensitivity is re-introduced whenever an extraction error maps an instance onto a form whose validity happens to align with plausibility. Reducing the overall content effect therefore requires tightening extraction—both the structured extractor used on the symbolic path and the direct LLM fallback—since the symbolic validator contributes no content-dependent errors but cannot correct them either.

5.3 Multi-Model Extraction

To test whether the design claim—that the symbolic stage is the invariant, content-immune component—holds across backbones, we re-run the pipeline with the symbolic validator, post-processing rules, and LLM fallback held fixed while swapping the extractor LLM. Table 6 reports results for two backbones on the full 191-instance test set.

Two observations are notable. First, combined score and TCE are essentially unchanged across backbones (56.57 vs. 56.47; TCE 1.02 vs. 1.06), while accuracy tracks extractor quality

Configuration	Acc.	TCE
Full system ($k = 3$)	97.0%	2.78
– self-consistency ($k = 1$)	94.0%	4.95
– I/O post-processing	96.0%	~ 4.0
– 24 $\rightarrow$ 15 valid forms	93.0%	~ 5.5
Baseline (initial prompt)	70.0%	> 10

Table 7: Ablation study on the training set (200 instances). “–” indicates removal of a component. “– 24 $\rightarrow$ 15 valid forms” replaces the Aristotelian set with its existential-import-free reduction.

(Claude Sonnet 4.5 reaches 97.38% vs. Qwen3-32B’s 96.34%). Content sensitivity, which the symbolic stage is designed to eliminate, remains near-zero regardless of backbone. Second, the stronger extractor reduces fallback frequency from 3.7% to 1.0%: Claude Sonnet 4.5 produces well-formed structured output on five instances where Qwen3-32B’s extraction failed, including several of the compound-noun-phrase cases identified in Section 6. Three of the five errors made by the Claude-based pipeline are on the same instances that Qwen3-32B also mis-classifies—specifically the Barbara figure-misidentification case and the E/O-confusion cases involving non-standard universal-negative phrasing (“do not overlap”; bare “are not”). These linguistic triggers are extractor-agnostic at current LLM capability levels, suggesting that further TCE gains require dedicated extractor training or stronger post-processing rather than backbone substitution.

5.4 Ablation Study

Table 7 reports the impact of key components on the training set (200 instances). Restoring the full 24 Aristotelian forms (existential import) and adding self-consistency voting yield the largest individual gains; I/O post-processing provides a modest but consistent improvement.

5.5 Paraphrase Probe

To probe sensitivity of extraction to surface form, we paraphrased 30 test instances (active $\leftrightarrow$ passive voice, synonym substitution for the middle term, and reordering of premises) and re-ran the full pipeline. Accuracy changes by **TBD_PP_DELTA** percentage points; all observed errors trace to the extraction stage. This confirms that the symbolic validator is invariant to surface form and that the extractor is the appropriate locus of future improvement.

6 Error Analysis

All 7 prediction errors on the test set are attributable to neural components of the pipeline; the symbolic validator introduces no errors on correctly extracted structures. Four errors occur on the symbolic path—from structured-extraction failures upstream of the validator—and three on the LLM fallback path, where the LLM makes a direct validity judgment without intermediate structure. The errors comprise 4 false negatives and 3 false positives. Per-error detail is given in Appendix C, Table 9.

The 7 failures cluster into three linguistic-trigger categories:

- **E/O type confusion (2/7):** errors on the symbolic path where non-standard universal-negative phrasing (“The set of X and Y do not overlap”; bare “X are not Y” without an explicit quantifier) is parsed as the particular O instead of the universal E. This turns valid forms such as Fresison (EIO-4) and Celaront (EAO-1) into invalid OIO-4 and OAO-1 respectively.
- **Figure or conclusion-scope error (2/7):** two symbolic-path errors where the structure of the argument is mis-identified. One is a failure on Barbara (AAA-1), the canonical valid syllogism, in which the middle term’s role is correctly identified but the figure is assigned as 3 instead of 1. The other is a conclusion whose scope (type O vs. type E) is genuinely ambiguous in surface form.
- **Compound noun phrase / extraction failure (3/7):** all three fallback errors, each triggered by the middle term appearing in syntactically different forms across premises (“stationery items” vs. “items of stationery”; “large body of saltwater” vs. “large bodies of saltwater”). In all three, structured extraction failed and the LLM fallback gave a content-consistent but logically incorrect verdict.

All three categories stress neural components of the pipeline. The symbolic validator is insensitive to surface form by construction: given correct types, terms, and figure, it would classify every instance correctly.

The 7 errors are distributed across plausibility conditions as (VP=2, VI=2, IP=1, II=2), a near-uniform spread that yields the low system-level TCE of 1.02.

Training-set annotation issues. During development we identified 5 annotation inconsistencies in the training set where universally valid Aristotelian forms (e.g. *Celarent* EAE-1, *Baroco* AOO-2) were labelled as invalid. These cases were communicated to the task organizers and do not appear in the test set. This observation is a positive side-effect of the interpretability of the neuro-symbolic paradigm: because every prediction is explainable by a mood-figure pair, labelling anomalies are easy to detect.

7 Discussion

The strict separation of neural extraction from symbolic validation offers three advantages: (1) the symbolic component is provably immune to content effects; (2) errors are localizable to either extraction or validation; (3) every prediction is interpretable via the extracted mood-figure pair.

Existential import. A critical design decision was adopting the Aristotelian interpretation with existential import (24 valid forms) rather than modern-logic (15 forms). Ablations with 15 forms showed substantially lower accuracy ($\sim 93\%$), confirming that the task adopts the traditional convention.

Where the residual content effect lives. In principle, a perfect symbolic validator achieves $TCE = 0$. The Section 5.2 decomposition shows that the symbolic validator itself contributes no content-dependent errors: all residual TCE comes from neural components, distributed between the extractor (4/7 errors) and the fallback (3/7 errors). Further lowering TCE therefore requires tightening neural components rather than changes to the validator.

Sampling is not enough. The B2→B3 comparison in Section 5.1 shows that adding self-consistency sampling on top of chain-of-thought yields no improvement in TCE (9.55 vs 10.59) and a marginal decrease in combined score. This is informative: when multiple samples share the same plausibility-driven bias, majority voting reinforces rather than cancels it. Reducing content effect in pure-LLM systems thus requires a structural intervention, not more compute.

Generalisation beyond categorical syllogisms.

The symbolic validator is tied to the 24 Aristotelian categorical forms. The decomposition itself, however, is form-agnostic: extending the sys-

tem to propositional or first-order fragments requires swapping the validator (e.g. a SAT/SMT backend) while preserving the extract-then-validate separation. The neural stage’s role—mapping natural language to a formal structure—remains the same.

Cross-model robustness. The multi-model results (Section 5.3) indicate that accuracy tracks extractor quality while combined score and TCE remain stable across backbones, consistent with the architectural claim that content-invariance is contributed by the symbolic stage rather than any particular LLM. Swapping Qwen3-32B for Claude Sonnet 4.5 also nearly eliminates fallback invocation (from 3.7% to 1.0%), which suggests that most of the Qwen-era fallback cases were extraction-difficulty artefacts rather than intrinsic problem difficulty.

8 Conclusion

We presented a neuro-symbolic system for SemEval-2026 Task 11 that achieves 96.34% accuracy, a TCE of 1.02, and a combined score of 56.57, demonstrating near-complete immunity to content effects. Controlled comparisons on a shared backbone show that reducing the LLM’s reasoning role—rather than adding sampling or reflexion on top of an LLM reasoner—is what closes the content-effect gap: direct prompting, chain-of-thought, and self-consistent chain-of-thought all plateau with combined scores in the 19–27 range, while our system reaches 56.57. A branch-level decomposition confirms that the symbolic validator contributes no content-dependent errors and localizes the residual content sensitivity to the neural components: the structured extractor (4/7 errors) and the LLM fallback (3/7 errors). Swapping the extractor LLM to Claude Sonnet 4.5 preserves combined score and TCE while reducing fallback frequency from 3.7% to 1.0%, supporting the claim that content-invariance is contributed by the symbolic stage rather than any particular backbone. Future work includes extending the validator to propositional and first-order fragments, reducing the fallback footprint, and adapting the pipeline to the multilingual and irrelevant-premise subtasks.

Limitations

Our system is evaluated only on Subtask 1 (English binary classification over categorical syllogisms). It does not address the multilingual setting

of Subtask 3, the irrelevant-premise conditions of Subtasks 2 and 4, or non-categorical logical fragments. The symbolic-path accuracy depends on the extractor LLM’s ability to parse noun phrases and quantifier scope; paraphrases and double negations remain the dominant extraction failure mode (Section 6). The LLM fallback, invoked on 3.7% of test instances with Qwen3-32B and 1.0% with Claude Sonnet 4.5, is the other neural component that can reintroduce content sensitivity; reducing fallback frequency by improving extraction is the most direct way to lower system-level TCE. Finally, our evaluation is limited to the 191 official test instances; a paraphrase probe on a 30-instance subsample (Section 5.5) provides only a preliminary indication of robustness to surface-form variation.

Ethical Considerations

This work uses the SemEval-2026 Task 11 dataset, which consists of synthetic categorical syllogisms in English with no personally identifying information. The system performs binary formal-validity classification and is not designed for user-facing deployment; we do not foresee direct risks of misuse. The extraction and fallback components rely on external LLMs (Qwen3-32B via Amazon Bedrock and Claude Sonnet 4.5 via the Anthropic API, for the multi-model comparison), whose training data we do not control; any biases in those models may influence extraction on rare linguistic constructions. By design, however, the symbolic validator is insensitive to such biases, and the branch-level decomposition (Section 5.2) makes the residual content sensitivity measurable.

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References

Leonardo Bertolazzi, Albert Gatt, and Raffaella Bernardi. 2024. A systematic analysis of large language models as soft reasoners: The case of syllogistic inferences. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language*

Processing (EMNLP). Association for Computational Linguistics.

Irving M. Copi, Carl Cohen, and Kenneth McMahon. 2014. *Introduction to Logic*, 14 edition. Pearson.

Ishita Dasgupta, Andrew K. Lampinen, Stephanie C. Y. Chan, Hannah R. Sheahan, Antonia Creswell, Dharshan Kumaran, James L. McClelland, and Felix Hill. 2022. Language models show human-like content effects on reasoning tasks. *arXiv preprint arXiv:2207.07051*.

Tiwalayo Eisape, Michael Henry Tessler, Ishita Dasgupta, Fei Sha, Sjoerd van Steenkiste, and Tal Linzen. 2024. A systematic comparison of syllogistic reasoning in humans and language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. Association for Computational Linguistics.

Geonhee Kim, Marco Valentino, and André Freitas. 2025. Reasoning circuits in language models: A mechanistic interpretation of syllogistic inference. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 10074–10095, Vienna, Austria. Association for Computational Linguistics.

Qing Lyu, Shreya Havaldar, Adam Stein, Li Zhang, Delip Rao, Eric Wong, Marianna Apidianaki, and Chris Callison-Burch. 2023. Faithful chain-of-thought reasoning. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AACL)*. Association for Computational Linguistics.

Giuseppe Maraia, Marco Valentino, Fabio Massimo Zanzotto, and Leonardo Ranaldi. 2026. Abstract activation spaces for content-invariant reasoning in large language models. *arXiv preprint arXiv:2602.02462*.

Kentaro Ozeki, Risako Ando, Takanobu Morishita, Hirohiko Abe, Koji Mineshima, and Mitsuhiro Okada. 2024. Exploring reasoning biases in large language models through syllogism: Insights from the NeuBAROCO dataset. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics.

Xin Quan, Marco Valentino, Louise A. Dennis, and André Freitas. 2024. Verification and refinement of natural language explanations through LLM-symbolic theorem proving. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics.

Leonardo Ranaldi, Marco Valentino, and André Freitas. 2025. Improving chain-of-thought reasoning via quasi-symbolic abstractions. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics.

S. M. Seals and Valerie L. Shalin. 2024. Evaluating the deductive competence of large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. Association for Computational Linguistics.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36.

Marco Valentino, Geonhee Kim, Dhairya Dalal, Zhixue Zhao, and André Freitas. 2025. Mitigating content effects on reasoning in language models through fine-grained activation steering. *arXiv preprint arXiv:2505.12189*.

Marco Valentino, Leonardo Ranaldi, Giulia Pucci, Federico Ranaldi, and André Freitas. 2026. SemEval-2026 Task 11: Disentangling content and formal reasoning in large language models. In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, San Diego, California. Association for Computational Linguistics.

Jundong Xu, Hao Fei, Liangming Pan, Qian Liu, Mong-Li Lee, and Wynne Hsu. 2024. Faithful logical reasoning via symbolic chain-of-thought. *arXiv preprint arXiv:2405.18357*.

A Extraction Prompt Template

The extraction prompt instructs the LLM to:

1. Identify the two premises and the conclusion.
2. For each proposition, determine the type (A, E, I, or O), paying special attention to the distinction between I (“Some S are P”) and O (“Some S are *not* P”).
3. Identify the three terms (S, P, M) based on their distribution: S appears in the minor premise and the conclusion; P appears in the major premise and the conclusion; M appears in both premises but not in the conclusion.
4. Output the proposition types for the two premises and the conclusion (e.g. “EIO”), together with the three terms in a structured format.

The syllogistic figure (1–4) is then computed deterministically from the position of M in the two premises:

- Figure 1: M is subject in P1, predicate in P2.
- Figure 2: M is predicate in both premises.
- Figure 3: M is subject in both premises.
- Figure 4: M is predicate in P1, subject in P2.

It.	Change	Acc.	Score
0	Baseline prompt	70%	20
1	Structured extraction	90%	33
2	Self-consistency voting	94%	33.8
3	I/O post-processing	96%	40
4	24 Aristotelian forms	97%	41.7
5	Compound-term handling	97.5%	45
6	Double-negation handling	98%	48

Table 8: Development progression on the training set (200 instances). Values are approximate.

#	Type	Sub.	Gold	Pred.	Path	Trigger
1	FN	VP	AAA-1 (Barbara)	AAA-3	sym	figure misid.
2	FN	VI	EIO-4 (Fresison)	OIO-4	sym	E/O confusion
3	FN	VI	EOA-1 (Celarent)	OAO-1	sym	E/O confusion
4	FP	II	AOE-2 [†]	AOO-2	sym	conclusion scope
5	FN	VP	— [‡]	extr. fail	fb	compound NP
6	FP	IP	— [‡]	extr. fail	fb	compound NP
7	FP	II	— [‡]	extr. fail	fb	compound NP

Table 9: All 7 test-set errors with dominant linguistic trigger (Qwen3-32B submitted run). FN=false negative (valid→invalid), FP=false positive (invalid→valid). Path: sym=symbolic, fb=fallback. Subgroups: VP/VI/IP/II as defined in §5. [†]Conclusion is genuinely ambiguous between O (yielding AOO-2, valid) and E (yielding AOE-2, not in the 24 valid forms); gold labels the instance invalid. [‡]Fallback cases have no extracted form; gold validity is *valid* for #5 and *invalid* for #6 and #7.

B Complete List of Development Iterations

Table 8 summarizes the progression of system performance through iterative development on the training set.

C Test Set Error Details

Table 9 lists all 7 errors on the official test set with their dominant linguistic trigger.

D Multi-Model Extractor Details

The multi-model study (Section 5.3) uses two backbones:

- **Qwen3-32B** (the submitted system’s extractor), accessed via Amazon Bedrock.
- **Claude Sonnet 4.5** (model ID `claude-sonnet-4-5-20250929`), accessed via the Anthropic API.

Both configurations use identical prompt templates, self-consistency sampling ($k = 3$ at temperatures $[0.1, 0.3, 0.5]$), post-processing rules, deterministic figure computation, symbolic validator, and LLM

fallback. Only the extractor LLM differs between the two rows of Table 6.